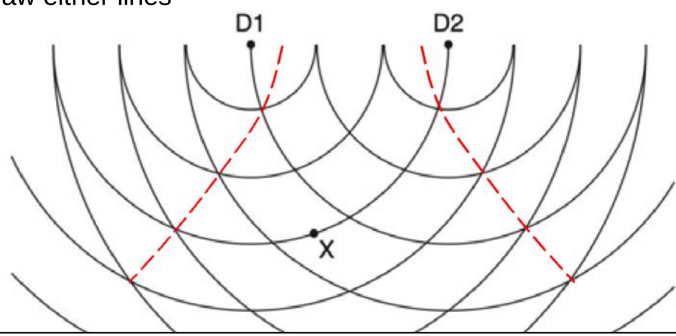
4	**(a)**	**(i)**	**1.**	<u>constant phase difference</u> (and does not vary with time) <u>between the waves</u> (produced by the dippers)	B1
			2.	connect dippers to same vibrator/motor	B1

		(ii)	overlapping waves have <u>similar / same amplitude</u>	B1
	(b)	any means of 'freezing' the pattern e.g. use of a stroboscope, video		B1
	(c)	evidence of using ≥ 2 cycles to calculate period (T = 0.21 s) or frequency $\lambda = \frac{v}{f} = vT = 0.40 \times 0.21$ so, $\lambda = 0.084$ m or 8.4 cm		B1 B1
	(d)	**(i)**	Evidence to deduce distances of X from source (e.g. X is 3λ and 3.5λ from D1 and D2, X on 3rd wavefront from D1 and between 3rd and 4th wavefronts from D1) path difference $= 3.5\lambda - 3.0\lambda = 0.5\lambda$ $= 4.2$ cm	M1 A1
		(ii)	180°	B1
	(e)	draw either lines 		B1

5	**(a)**	(electrostatic) force between two point charges proportional to product of charges and inversely	